Matrices

Linear Algebra

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Definition 1 (Matrix (real))

Let m e n be two positive integers. A real matrix with m rows and n columns, is a table with $m \times n$ real numbers arranged in m rows and n columns.

☞ *We will use the following notation*

$$\mathrm{A} = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n-1} & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n-1} & a_{2,n} \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n-1} & a_{m,n} \end{bmatrix}$$

where sub indices of each entry $a_{i,j}$ means that $a_{i,j}$ is located in i-th row and j-th column of the matrix A.
Or in the compact notation

$$\mathrm{A} = [a_{i,j}]_{\substack{1 \le i \le m \\ 1 \le j \le n}}.$$

- A matrix with m rows and n columns it is said a matrix of size $m \times n$. The set of all real matrices of size $m \times n$ will be represented by the symbol $\mathscr{M}_{m,n}$.
- A matrix $A \in \mathscr{M}_{m,n}$ is said to be **rectangular** if $m \neq n$. As particular cases we have:
  - *column matrices* $(n = 1)$

$$c = \begin{bmatrix} c_{1,1} \\ c_{2,1} \\ \vdots \\ c_{m,1} \end{bmatrix}$$

  - and *row matrices* (m=1)

$$l = \begin{bmatrix} l_{1,1} & l_{1,2} & \cdots & l_{1,n} \end{bmatrix}$$

- A matrix $A \in \mathscr{M}_{m,n}$ is **square** if $m = n$. For simplicity, we say that a square matrix (of size $n \times n$) has **order** n and we write $A \in \mathscr{M}_n$.
  - the **main diagonal** of a matrix A is te set of entries $a_{i,j}$ such that $i = j$, $i = 1, 2, \ldots, n$
  - the **secondary diagonal** of a matrix A is te set of the entries $a_{i,j}$ such that $i + j = n + 1$.
  - Given a matrix A) $\in \mathscr{M}_{m,n} = [a_{i,j}]_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n}}$ (not necessarily square), we define **main diagonal** of the matrix A as being the set of the entries $\{a_{1,1}, a_{2,2}, \ldots, a_{\min(m,n),\min(m,n)}\}$.
  - The secondary diagonal is not defined for rectangular matrices.

Examples: Entries that belong to main diagonals are colored in red and the entries that belong secondary diagonal are colored in blue azul.

$$a)\ \begin{bmatrix} \pi & 1 & 2 & -1 \\ 2 & \sqrt{2} & 0 & \pi \\ 1 & \sqrt{3} & 1 & \pi \\ 0 & -3 & \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \qquad b)\ \begin{bmatrix} \pi & 1 & 2 & 8 \\ 1 & 3 & 4 & -3 \\ 1 & -1 & 3 & -3 \end{bmatrix} \qquad c)\ \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 3 & 3 \\ 4 & 4 \end{bmatrix}$$

- A square matrix $[a_{i,j}]_{\substack{1\leq i\leq n \\ 1\leq j\leq n}}$, of order n is said to be **upper triangular** if and only if $i > j \Rightarrow a_{i,j} = 0$.
- A square matrix $[a_{i,j}]_{\substack{1\leq i\leq n \\ 1\leq j\leq n}}$, of order n is said to be **lower triangular** if and only if $i < j \Rightarrow a_{i,j} = 0$.

- A square matrix $[a_{i,j}]_{\substack{1\leq i\leq n\\ 1\leq j\leq n}}$, of order n, is said to be **diagonal** if is lower and upper triangular ($a_{i,j}=0$ para todo $i\neq j$).
- If a diagonal matrix has all entries of main diagonal equals is said to be a **escalar** matrix.

☞ *If the entries of the main diagonal of a scalar matrix are equal to* 1 *we said that the matrix is the* ***identity matrix*** *of order n and we denoted by* $\mathsf{I_n}$.

Examples:

$$\mathsf{D}=\begin{bmatrix}1&0&0\\0&\sqrt{5}&0\\0&0&-2\end{bmatrix},\quad \mathsf{E}=\begin{bmatrix}\frac{1}{2}&0&0&0\\0&\frac{1}{2}&0&0\\0&0&\frac{1}{2}&0\\0&0&0&\frac{1}{2}\end{bmatrix},\quad \mathsf{I_4}=\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix}$$

Matrices D, E e $\mathsf{I_4}$ are all diagonal matrices. Matrizes E e $\mathsf{I_4}$ are scalar matrices, being that $\mathsf{I_4}$ is the identity matrix of order 4.

- Two matrices A e B are said to be **equal**, $A = B$, if they have same size and all corresponding entries are equal, i. e. $a_{i,j} = b_{i,j}$ for all i and j. Otherwise, A e B are said to be **different** ($A \neq B$).
- Let $A \in \mathscr{M}_{m,n}$ One matrix A' is said to be a **submatrix** of a matrix A, if A' is obtained from A by removing any number of rows or columns of A

Definition 2 (Addition of matrices)

Let $A = [a_{i,j}]_{\substack{1\leq i\leq m\\ 1\leq j\leq n}}$ and $B = [b_{i,j}]_{\substack{1\leq i\leq m\\ 1\leq j\leq n}}$ two matrices, of same size $(m \times n)$.
We define the *sum* of matrix A with matrix B as being a matrix C of size $m \times n$, whose entries ,$c_{i,j}$, are determined by

$$c_{i,j} = a_{i,j} + b_{i,j},$$

and we write, $C = A + B$

☞ *The addition of two matrices with different size is not defined.*

Examples: If,

$$A = \begin{bmatrix} x & 0 & 1 \\ -1 & -1 & y \\ 0 & 2 & -1 \end{bmatrix}; \quad B = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}; \quad C = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} x+2 & -9 & y \\ -1 & -z & y \\ 0 & -1 & -x \end{bmatrix}$$

Than:

$$A + D = \begin{bmatrix} 2x+2 & -9 & 1+y \\ -2 & -(1+z) & 2y \\ 0 & 1 & -(1+x) \end{bmatrix}; \quad C + B = \begin{bmatrix} a-1 \\ b \\ c+1 \end{bmatrix}$$

$$A + I_3 = \begin{bmatrix} x+1 & 0 & 1 \\ -1 & 0 & y \\ 0 & 2 & 0 \end{bmatrix}$$

Properties 1 (of addition of matrices)

1. The addition is a *closed operation* in the set $\mathscr{M}_{m,n}$

$$\forall A, B \in \mathscr{M}_{m,n} \ , \ \exists C \in \mathscr{M}_{m,n} \ : \ A + B = C.$$

2. The addition is an *associative operation*

$$\forall A, B, C \in \mathscr{M}_{m,n} \ , \ (A + B) + C = A + (B + C).$$

3. Exists a neutral element, E, in $\mathscr{M}_{m,n}$.

$$\exists E \in \mathscr{M}_{m,n} \ , \ \forall A \in \mathscr{M}_{m,n} \ : \ A + E = E + A = A.$$

4. All matrix A in $\mathscr{M}_{m,n}$, has *additive inverse* A'

$$\forall A \in \mathscr{M}_{m,n} \ , \ \exists A' \in \mathscr{M}_{m,n} : \ A + A' = A' + A = E.$$

5. The addition of matrices is a *commutative operation*,

$$\forall A, B \in \mathscr{M}_{m,n} \ : \ A + B = B + A.$$

- The neutral element $\mathscr{M}_{m,n}$ is the matrix $\mathrm{E} = 0_{m,n}$, where $0_{m,n}$ is the matrix of size $m \times n$ whose all entries are zero. We call the matrix $0_{m,n}$ the **zero or null matrix of size $m \times n$.**
- **The additive inverse of matrix $\mathrm{A} = [a_{i,j}]_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n}}$ is denoted by $-\mathrm{A}$ and we have $-\mathrm{A} = [-a_{i,j}]_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n}}$.**

Definition 3 (subtraction of matrices)

Given two matrices $\mathrm{A}, \mathrm{B} \in \mathscr{M}_{m,n}$ we define the subtraction $\mathrm{A} - \mathrm{B}$ as being the addition of A with additive inverse of B.

Definition 4 (Multiplication of a real number by a matrix)

Let α a real number and $A = [a_{i,j}]_{\substack{1\leq i\leq m\\ 1\leq j\leq n}}$ a matrix of size $m \times n$. The *scalar multiplication* of α by matrix A is a matrix of same size of A, whose entries are equal to the correspondent entries of matrix A multiplied by α.

$$\alpha A = [\alpha a_{i,j}]_{\substack{1\leq i\leq m\\ 1\leq j\leq n}}.$$

Examples:

$$\pi \begin{bmatrix} \pi & -1 & \alpha \\ 2 & \pi^{-1} & -2 \end{bmatrix} = \begin{bmatrix} \pi^2 & -\pi & \pi\alpha \\ 2\pi & 1 & -2\pi \end{bmatrix}$$

$$3 \begin{bmatrix} -1 & 3 & -1/3 \\ 0 & 1 & 2 \\ 4 & -2 & 0 \end{bmatrix} = \begin{bmatrix} -3 & 9 & -1 \\ 0 & 3 & 6 \\ 12 & -6 & 0 \end{bmatrix}$$

Properties 2

For all matrices A and B in $\mathscr{M}_{m,n}$ and for all real numbers α and β, we have

1. $\alpha(A+B) = \alpha A + \alpha B$.
2. $(\alpha+\beta)A = \alpha A + \beta A$.
3. $(\alpha\beta)A = \alpha(\beta A)$.
4. $1A = A$.
5. $(-1)A = -A$.

Definition 5 (Multiplication of matrices)

Consider the matrices $A = [a_{i,j}]_{\substack{1\leq i\leq m\\ 1\leq j\leq \ell}} \in \mathscr{M}_{m,\ell}$ e $B = [b_{i,j}]_{\substack{1\leq i\leq \ell\\ 1\leq j\leq n}} \in \mathscr{M}_{\ell,n}$. We define the *product* of matrix A by matrix B as being the matrix $C = [c_{i,j}]_{\substack{1\leq i\leq m\\ 1\leq j\leq n}} \in \mathscr{M}_{m,n}$, and we write $C = A \times B$, or $C = AB$, being the entries of matrix C given by the relation

$$c_{i,j} = \sum_{k=1}^{\ell} a_{i,k}\, b_{k,j}.$$

☞ *The multiplication of* $A \times B$ *is only defined if the number of columns of left matrix* A*, is equal to the number os rows of right matrix* B*.*

Example:

$$AB = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 0 & 3 \\ 1 & -1 \\ 2 & 0 \end{bmatrix} =$$

$$= \begin{bmatrix} 1\times 0 + 2\times 1 + 3\times 2 & 1\times 3 + 2\times(-1) + 3\times 0 \end{bmatrix} = \begin{bmatrix} 8 & 1 \end{bmatrix}.$$

☞ *The multiplication* BA **is not defined**, *as the left matrix has* 2 *columns and the right matrix has* 1 *raw.*

☞

- *If the matrices* A *e* B *are square matrices with same size, than the multiplications* AB *e* BA *are both defined.*
- *The multiplication of matrices is not a commutative operation. This means, the proposition*

$$\forall A, B \in \mathscr{M}_n,\ AB = BA$$

is false.

Definition 6 (Commuting matrices)

We say that two matrices $A, B \in \mathscr{M}_n$ are *commuting matrices* if and only if is valid the equality

$$AB = BA.$$

Definition 7 (Center of $\mathscr{M}_n$)

The *center* of $\mathscr{M}_n$ is the set of all square matrices of size n that commute with all square matrices of size n and is denoted by $\mathscr{Z}$.

$$\mathscr{Z}(\mathscr{M}_n) = \left\{ A \in \mathscr{M}_n : AX = XA, \ \forall X \in \mathscr{M}_n \right\}.$$

Examples:

Consider the following square matrices of size 2:

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix},\ B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix},\ C = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} \text{ e } D = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}.$$

1. A do not commute with B because,
$$AB = \begin{bmatrix} -1 & -1 \\ 2 & 2 \end{bmatrix} \text{ and } BA = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$
2. A commute with C because,
$$AC = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix} \text{ and } CA = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix}$$
3. The matrix $A \notin \mathcal{Z}(\mathcal{M}_2)$. In fact, if A belonged to the center of $\mathcal{M}_2$, it would have commute with all square matrices of size 2 and we saw that it do not comute with matrix B.
4. The matrix D commute with all square matrices of size2. In fact we have,
$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x & y \\ z & t \end{bmatrix} = \begin{bmatrix} x & y \\ z & t \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2x & 2y \\ 2z & 2t \end{bmatrix}$$
Than $D \in \mathcal{Z}(\mathcal{M}_2)$.

Properties 3

Consider the following matrices A, $B \in \mathscr{M}_{m,\ell}$, C, $D \in \mathscr{M}_{\ell,p}$ and $E \in \mathscr{M}_{p,n}$ and the scalar $\alpha \in \mathbb{R}$. The following properties are fullfied:

1. $(A+B)D = AD + BD$
2. $A(C+D) = AC + AD$
3. $(AD)E = A(DE)$
4. $0_{r,m}A = 0_{r,\ell}$ e $A0_{\ell,s} = 0_{m,s}$
5. $I_m A = A$ e $AI_\ell = A$
6. $\alpha(AC) = (\alpha A)C = A(\alpha C)$

Notes: First property is called right distributive property . The second is called left distributive property. The third is the associative law. The properties in 4, are respectively called, existence of absorbing element to the right and to the left. The properties in 5, are respectively called, existence of left and right identity element.

Definition 8 (Invertible Matrices)

We say that a matrix is invertible $A \in \mathscr{M}_n$ é *invertível* if exists a matrix $Z \in \mathscr{M}_n$ such that

$$A\,Z = Z\,A = I_n \tag{1}$$

If does not exists a matrix $Z \in \mathscr{M}_n$ such that (1) Than we say that matrix A is *not invertible*.



If a matrix A is invertible than exists only one matrix Z that fulfill the relation (1), and we call to matrix Z the inverse of A. Frequently we denote the inverse of a matrix A by A^{-1}.

Example: The matrix

$$A = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}$$

is invertible, because

$$\begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1/3 & -1/3 \\ 1/3 & 2/3 \end{bmatrix} = \begin{bmatrix} 1/3 & -1/3 \\ 1/3 & 2/3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

than, we have

$$A^{-1} = \begin{bmatrix} 1/3 & -1/3 \\ 1/3 & 2/3 \end{bmatrix}.$$

and the inverse matrix of matrix A^{-1} é a matriz A. Ou seja, tem-se $(A^{-1})^{-1} = A$.

Definition 9 (Power of order $k \in \mathbb{Z}$)

Let $A \in \mathscr{M}_n$ and k an integer. The *power* of order k of the matrix A is the matrix $A^k \in \mathscr{M}_n$ where,

$$A^0 = I_n$$

$$A^k = \underbrace{A \times A \times \ldots \times A}_{k \text{ vezes}} \quad \text{se } k > 0.$$

$$A^{-k} = \underbrace{A^{-1} \times A^{-1} \times \ldots \times A^{-1}}_{k \text{ vezes}} \quad \text{se } k > 0 \quad \text{(A invertível)}.$$

Definition 10 (Transpose (of a matrix))

Let $A = [a_{i,j}]_{\substack{1 \le i \le m \\ 1 \le j \le n}}$ a matrix of size $m \times n$. The *transpose* of matrix A is a matrix of size $n \times m$, and is denoted by $A^T = [a'_{i,j}]_{\substack{1 \le i \le n \\ 1 \le j \le m}}$, where

$$a'_{i,j} = a_{j,i}, \quad 1 \le i \le n \text{ e } 1 \le j \le m.$$

Examples: Consider the matrices

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix} \quad \text{e} \quad C = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Than,

$$A^T = \begin{bmatrix} 1 & 5 \\ 2 & 6 \\ 3 & 7 \\ 4 & 8 \end{bmatrix} \quad B^T = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix} \quad \text{e} \quad C^T = \begin{bmatrix} -1 & 0 & 1 \end{bmatrix}.$$

$$(A^T)^T = ?, \quad (B^T)^T = ?, \quad (C^T)^T = ?$$

Definition 11 (Symmetric and Skew-Symmetric Matrices)

Let A be a square matrix.

- We say that A is a **symmetric matrix** if $A = A^T$.
- If $A = -A^T$, we say that A is a **skew-symmetric matrix** .

Examples:

1. The symmetric matrices of order 3 have the following general form

$$\begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix}, \quad a, b, c, d, e, f \in \mathbb{R}$$

2. The skew-symmetric matrices of order 3 have the following general form

$$\begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix}, \quad a, b, c \in \mathbb{R}.$$

Properties 4

1. $(A^T)^T = A$
2. $(A \pm B)^T = A^T \pm B^T$
3. $(AC)^T = C^T A^T$
4. $(A^k)^T = (A^T)^k$, $k \in \mathbb{N}_0$
5. $(BC)^{-1} = C^{-1} B^{-1}$
6. $(B^k)^T = (B^T)^k$, $k \in \mathbb{Z}$

☞ *The properties are valid for all matrices, as long as the indicated operations make sense.*

Definition 12 (Leading coefficient)

Let E be a matrix. For each row of E the leading coefficient is the first nonzero entry in that row. If a row contains only null entries it does not have leading coefficient.

Definition 13 (Row echelon form (r.e.f.))

We say that a matrix E is in row echelon form if:

1. For all row $i > 1$, of E, the leading coefficient is to the right of the one above it.
2. Any non-zero rows are always above rows with all zeros.

Examples: Consider the following matrices

$$A = \begin{bmatrix} 0 & 1 \\ 2 & 0 \\ 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 4 \end{bmatrix}.$$

- The only matrix that is in row echelon form is matrix B.
- In matrix A the leading coefficient of the second row is left of the leading coefficient of first row .
- In matrix C the leading coefficient of the 3-th row is not right of the leading coefficient of the second row (they are on same column).
- The matrix D has a non-zero row below of a zero row.

Definition 14 (Rank of a matrix that is in row echelon form)

The rank of a matrix E that is in row echelon form , *rank*(E) is the number of non-zero rows.

$$\boxed{r(E) = \text{number of non-zero rows of E.}}$$

☞ ***Note that this definition is only valid for matrices that are on row echelon form.***

We will see below hot to compute the rank of any matrix.

Definition 15 (Elementary row (column) operations)

Let A be a matrix. The following operations are called *elementary row (column) operation* :

1. Row swap, exchange any two rows of A. We will denote the exchange of i-th with j-th rows, L_i and L_j, (of columns C_i e C_j) by $L_i \leftrightarrow L_j$ ($C_i \leftrightarrow C_j$).
2. Multiply a row (column) L_i (C_i) by a non-zero scalar αL_i (coluna βC_i), onde $\alpha, \beta \in \mathbb{R} \setminus \{0\}$. This elementar operation will be denoted by,

$$L_i \leftarrow \alpha L_i \quad (\ C_i \leftarrow \beta C_i \ )$$

3. Add a multiple of a row L_j (column C_j) to another row L_i (column C_i). This elementar operation will be denoted by,

$$L_i \leftarrow L_i + \alpha L_j \quad (\ C_i \leftarrow C_i + \beta C_j \ )$$

Definition 16 (Equivalent Matrices)

We say that two matrices are **equivalents** if one matrix is obtained from the other by applying a finite number of elementary row operations.

Theorem 1

Let A be a matrix equivalent to a matrix E that is in echelon form. Than we have,

$$\boxed{\text{rank}(A) = \text{rank}(E)}$$

☞ *This theorem allows to compute the rank of any matrix since we find a matrix* E *equivalent to* E *and in echelon form –> Gauss elimination algorithm.*

A first example (Gaussian algorithm): Find a echelon form equivalent to matrix A.

$$\overbrace{\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 2 & 3 & 1 \end{bmatrix}}^{\mathrm{A}} \xrightarrow{L_1 \leftrightarrow L_2} \overbrace{\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 2 & 3 & 1 \end{bmatrix}}^{\mathrm{A}_1} \xrightarrow{L_4 \leftarrow L_4 + (-1)L_1} \overbrace{\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 2 & 2 & 0 \end{bmatrix}}^{\mathrm{A}_2} \rightarrow$$

$$\xrightarrow{L_4 \leftarrow L_4 + (-2)L_2} \overbrace{\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix}}^{\mathrm{A}_3} \xrightarrow{L_3 \leftrightarrow L_4} \overbrace{\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}}^{\mathrm{E}}$$

matrices A, A_1, A_2, A_3 and E are all equivalents than they have equal rank. The matrix E is in echelon form. As E has 3 non-zero rows we have,

$$r(\mathrm{E}) = r(\mathrm{A}_3) = r(\mathrm{A}_2) = r(\mathrm{A}_1) = r(\mathrm{A}) = 3.$$

Algorithm 1 (Gaussian elimination)

Consider the matrix $A \in \mathscr{M}_{m,n}$.

1. Let $U = A$ and $r = 1$. if $U = 0$, U is in row echelon form (**STOP**).
2. If $U \neq 0$, search in rows with indices $r, \ldots, m$, of U, the first non-zero column, k, and first entry $a_{i,k} \neq 0$. If $i > r$, swap rows r e i in U (we get a non-zero entry in na (r,k) position). Let U be the matrix obtained by this swap.
3. Add multiples of r-th row to the below rows to vanish all entries in the k-th column and rows with indices $r+1, \ldots, m$. Let matrix U be the resulted matrix.
4. If $r = m-1$ or the rows $r+1, \ldots, m$ are all zero, U is in row echelon form(**STOP**). Otherwise, make $r = r+1$ and repeat steps 2, 3 e 4.

☞ *In step 3, we did not explained which elementar operations allow to vanish the entries on k-th column and rows with indices $r+1, \ldots, k$. It is easy to verify that we vanish the entry $a_{\ell,k}$ by applying the operation $L_\ell \leftarrow L_\ell - m_\ell L_r$, com $m_\ell = a_{\ell,k}/a_{r,k}$.*

Example: Gaussian elimination applied to matrix A.

$$A = \begin{bmatrix} 2 & 6 & 4 & 4 \\ -4 & -12 & -8 & -7 \\ 0 & 0 & -1 & -4 \\ 1 & 3 & 1 & -2 \end{bmatrix}.$$

Step 1. $U^{(1)} = A$ e $r = 1$.
Step 2. As $a_{1,1} \neq 0$, is not necessary to swap rows.
Step 3. Vanish the entries in first column and below $a_{1,1}$. We have,

$$m_2 = \frac{-4}{2}, \quad m_3 = \frac{0}{2}, \quad \text{e} \quad m_4 = \frac{1}{2},$$

Then, we must apply the following operations $L_2 \leftarrow L_2 - (-2)L_1$ e $L_4 \leftarrow L_4 - \frac{1}{2}L_1$. And we get the matrix

$$U^{(2)} = \begin{bmatrix} 2 & 6 & 4 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & -4 \\ 0 & 0 & -1 & -4 \end{bmatrix}.$$

Step. The sub-matrix $U^{(2)}$, constituted by rows with indices 2, 3 e 4 is not the null matrix, then we increment r, to $r = 2$ and we repeat step 2.

example (cont.)

Step 2. The first non-zero column, of the sub-matrix $U^{(2)}$, constituted by rows with indices 2, 3 e 4, is the 3-th and the first no zero entry is in 3-th row of $U^{(2)}$. Then we swap rows $L_2 \leftrightarrow L_3$ in $U^{(2)}$ and we get

$$U^{(3)} = \begin{bmatrix} 2 & 6 & 4 & 4 \\ 0 & 0 & -1 & -4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & -4 \end{bmatrix}.$$

Step 3. Vanish entries of 3-th column below entry $a_{2,3}$. We have $m_4 = -1/-1$ and we apply the operation $L_4 \leftarrow L_4 - L_2$ to get the matrix

$$U^{(4)} = \begin{bmatrix} 2 & 6 & 4 & 4 \\ 0 & 0 & -1 & -4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Passo 4. The matix $U^{(4)}$ is in row echelon form and the Gaussian algorithm STOPs.

example (cont.)

We write:

$$\begin{bmatrix} 2 & 6 & 4 & 4 \\ -4 & -12 & -8 & -7 \\ 0 & 0 & -1 & -4 \\ 1 & 3 & 1 & -2 \end{bmatrix} \xrightarrow[L_4 \leftarrow L_4 - 1/2 L_4]{L_2 \leftarrow L_2 + 2L_1} \begin{bmatrix} 2 & 6 & 4 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & -4 \\ 0 & 0 & -1 & -4 \end{bmatrix} \xrightarrow{L_2 \leftrightarrow L_3} \begin{bmatrix} 2 & 6 & 4 & 4 \\ 0 & 0 & -1 & -4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & -4 \end{bmatrix}$$

$$\xrightarrow{L_4 \leftarrow L_4 - L_2} \begin{bmatrix} 2 & 6 & 4 & 4 \\ 0 & 0 & -1 & -4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

? Rank of A?

